

Log Analysis with SIEM

Centralisation
Correlation
Historic Event



SIEM

Host-Based Log sources → come from individual devices within the organisation

Network-Based Log Sources → collected data from network devices
IDS, IPS, firewalls, routers

web-based log sources → through web vulnerability

Important:

Time Pitfalls : different time zones

Log Normalisation

Alert Triage with Splunk

Initial Access Alert

Q1) How many failed login attempts were made on the user john.smith?

10.10.242.248 user = "john.smith" * Failed password *

Q2) What was the duration of the brute force attack in minutes?

10.10.242.248 user = "john.smith" * Failed password *

| stats earliest(-time) as attack-start latest(-time) as attack-end count as attempts

| eval duration - minutes = round((attack-end - attack-start)/60)

| convert ctime(~~start~~ attack-start) ctime(attack-end)

— convert ctime(attack-start) gives us on what time it started

Q3) What username was the attacker able to privilege escalate to?

~~some~~ index = "linux-alert" sourcetype = "linux-secure"
johnsmith (sudo OR su)

Q4) what is the name of the user account created by the attacker persistence?

index = "linux-alert" sourcetype = "linux-secure" root

Persistence Alert

1) What is the ProcessId of the process that created this malicious task?

index = "win-alert" host = WIN-H015 AssessmentTaskOne

2) What is the name of the parent process for the process that created this malicious task?

index = "win-alert" host = WIN-H015 AssessmentTaskOne
ParentProcessId = 4128

3) Which local group did the attacker enumerate during discovery?

index = "win-alert" oliver.thompson (EventCode = 4799 OR "net group" OR "net localgroup")

4) What is the name of the workstation from which the Threat Actor logged into this host?

index = "win-alert" oliver.thompson EventCode = 4624

Web Shell Alert

1) What time did the Brute-force activity using Hydra Begin?

index: "web-alert" 171.251.232.40 useragent="Mozilla/5.0
(Hydra) / sort + -time

2) Which user agent did the attacker use when interacting with the web shell?

index: "web-alert" 171.251.232.40 8374x.php

3) What was the number of requests made by the attacker to the server via the web shell?

index: "web-alert" 171.251.232.40 8374x.php method = POST

Alert Triage with Elastic

Investigating web Attacks

Q1) How many POST requests did the IP address 203.0.113.55 make to proxyLogon.ecp?

Q2) Which user-agent paired with the IP address 203.0.113.55 made the POST requests?

- index: weblogs and client.ip: "203.0.113.55" and http.request.method: "POST" and proxyLogon.ecp

Q3) How many logs contain the cmd= query parameter in the url.path field?

- index: weblogs and client.ip: "203.0.113.55" and http.request.method: 'GET' and cmd

Uncovering Account Activity

Q1) What is the winlog.record-id of the Administrator 4624 logon event?

@ timestamp >= "2025-07-20T05:10:00" and win.event-id: 4624 and winlog.event-data.TargetUserName: "Administrator"

Q2) What is the process pid of the Sysmon 1 event that occurred on Jul 20, 2025 @ 05:11:27.996?

@timestamp >= "2025-07-20T05:11:27" and winlog.event_id: "1"

Q3) What is the winlog_event_id for the new user account being created?

@timestamp >= "2025-07-20T05:11:27" and winlog.channel: "security" and winlog.task: "User Account Management"

Exposing Command Execution

Q1) What command does the attacker use to add the new account to the "Remote Desktop Users" group?

@timestamp >= "2025-07-20T05:13:15" and process.parent.name: "cmd.exe" and user.name: "Administrator"

Q2) What is the winlog.record-id of the 4732 Security event when attacker adds the user to the Administrator group?

@timestamp >= "2025-07-20T05:13:15" and winlog.event_id: "4732"

Q3) What Powershell command did the attacker run on Jul 20, 2025 @ 05:16:14.628?

@timestamp >= "2025-07-20T05:16:14" and event.module: "powershell"

Q4) what is the name of the archive that the attacker creates using the Rar.exe executable?

process.name: "Rar.exe"